

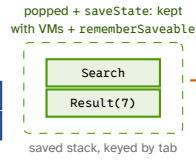
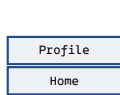
**In one line:** A NavController owns a **back stack of NavBackStackEntry** — each entry is its own **Lifecycle, ViewModelStore** and **SavedState** owner — and NavHost shows the top one. Routes are @Serializable classes (2.8+); every navigate() is a push plus **NavOptions** that pop, dedupe, save and restore. **Navigation 3** drops the controller: *you* own the back stack as a state list.

Bottom nav: navigate(tab) { popUpTo(graph.findStartDestination().id) { saveState = true }; launchSingleTop = true; restoreState = true }

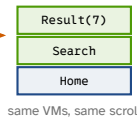
A · on Search, drilled in



B · on Profile



C · Search restored

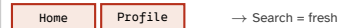


Drop one option (from A, tap Profile):

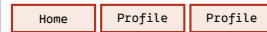
no popUpTo: stack grows, Back walks the tabs



popUpTo but no save/restore: Result(7) + VM gone



no launchSingleTop: re-tap Profile



## The model

- rememberNavController() + NavHost(nav, startDestination = Home) { ... } builds a NavGraph; currentBackStackEntryAsState() drives the bottom bar. Screens get **lambdas**, not the NavController (testable, previewable).
- Type-safe routes** (2.8.0, Sep 2024): @Serializable object/data class, fields = arguments (Bundle-encoded: pass **ids**; custom types need a typeMap). Read: entry.toRoute<Profile>(), in the VM savedInstanceState.toRoute<Profile>(). 2.9: KClass overloads, value classes; **2.10** (Aug 2026): KMP targets, predictive-pop transitions.
- navigation<Checkout>(startDestination = Cart) { ... } = nested graph: a flow with its own start, poppable as a unit (popUpTo<Checkout> { inclusive = true }).

## Example

```
@Serializable object Home
@Serializable data class Profile(val id: String)
@Serializable object Checkout; @Serializable object Cart
NavHost(nav, startDestination = Home) {
    composable<Home> { HomeScreen(onUser = { nav.navigate(Profile(it)) }) }
    composable<Profile>(deeplinks = listOf(navDeeplink<Profile>{
        basePath = "https://example.com/u" } { ProfileScreen() }
    navigation<Checkout>(startDestination = Cart) {
        composable<Cart> { entry ->
            val flow = remember(entry) { nav.getBackStackEntry<Checkout>() }
            val vm: CheckoutVM = hiltViewModel(flow) // shared by the flow
            CartScreen(vm, onDone = { nav.navigate(Home) {
                popUpTo<Checkout> { inclusive = true } } }) } } } }
```

## Back-stack operations

- popUpTo(X) pops all above X; inclusive = true pops X too (after login). launchSingleTop: no second copy on top.
- saveState/restoreState = **multiple back stacks** (2.4+): popped entries keep state + VMs, keyed by the tab.
- popBackStack() returns false if nothing popped (empty host = blank screen); navigateUp() goes to the parent inside your own task.

## Traps

- ! navigate() in the composable body runs on every recomposition — call it from an event or a LaunchedEffect.
- ! Double tap = two pushes: wrap in dropUnlessResumed { ... }.
- ! A whole object in a route: Bundle size, stale copy — pass the id.

**Remember:** entry = lifecycle + VM store + saved state · pop, single-top, save, restore · Nav 3: the stack is your list.

## ViewModel scope & results

- Default hiltViewModel() = the **entry**: survives rotation, cleared when popped. Graph-scoped (Example) lives while any of the graph is on the stack.
- Result back:** previousBackStackEntry?.savedStateHandle?.set("k", id); the caller collects its own savedInstanceState.getStateFlow("k", null). Or a shared graph VM — never a callback into a popped screen.

## Deep links & App Links

- <intent-filter> VIEW + DEFAULT + BROWSABLE + <data>; navDeepLink matches. The controller builds a **synthetic back stack** (start of each parent graph) so Back works; with FLAG\_ACTIVITY\_NEW\_TASK the task is cleared and rebuilt.
- App Links:** autoVerify="true" + /.well-known/assetlinks.json (package + SHA-256 cert). Android 12+: unverified links open the **browser**. adb shell pm get-app-links <pkg>.

## Predictive back

13: opt-in enableOnBackPressedCallback="true"; 15: system back-to-home / cross-activity animations for opted-in apps; **16 + targetSdk 36:** on by default — onBackPressed() / KEYCODE\_BACK no longer delivered. Use onBackPressedCallback, BackHandler, PredictiveBackHandler (a Flow<BackEventCompat> of progress); NavHost animates the pop (2.8+).

## Navigation 3 — you own the stack

**1.0.0 stable 19 Nov 2025** (1.2.0 Sep 2026), runtime KMP. rememberNavBackStack(Home) — keys @Serializable : NavKey, survives process death; push backStack.add(Profile(id)), pop removeLastOrNull(). NavDisplay(backStack, onBack, entryDecorators, entryProvider { entry<Home> { ... } }); per-entry VMs need rememberViewModelStoreNavEntryDecorator() (lifecycle-viewmodel-navigation3). SceneStrategy: several entries at once (list-detail, dialog). Tabs = lists you keep; ≈ SwiftUI NavigationStack(path:).